Identifying plasma modes in the ISM using synchrotron polarization.

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Plasma in the interstellar medium (ISM)

- is multiphase
  - Molecular clouds
  - Neutral media (cold, warm)
  - Ionized media (warm, hot)
- is magnetized
  - Linear polarization of synchrotron emission
  - Faraday rotation
- is turbulent
  - Multi-scale energy spectra
  - Non-thermal line broadening
  - Intermittency in velocity / density

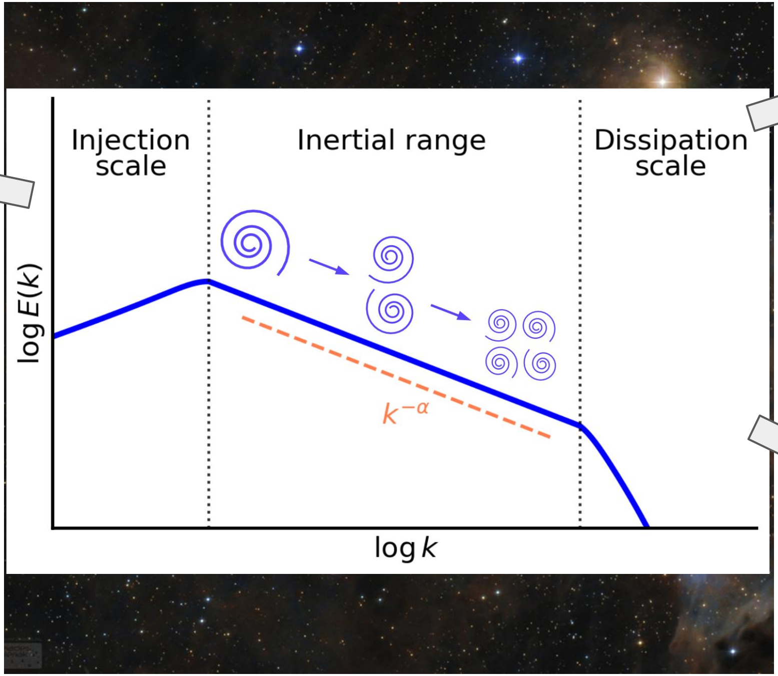
Image: APOD 2017 January 14, Lóránd Fényes

Intro

At larger scales

- Thermal phase exchanges
- Energy equipartition

(Draine2003, Heiles+2003, Hirashita+2010)



At inertial scales

- Heat regulation
- Opposing cloud collapse
- Gas fragmentation

(McKee+2007, Crutcher+2012, Fissel+2016)

At kinetic scales

- Dust-grain dynamics
- Particle acceleration
- CR scattering, transport

(Schlickeiser2000, Lemoine+2024, Gong+2023)

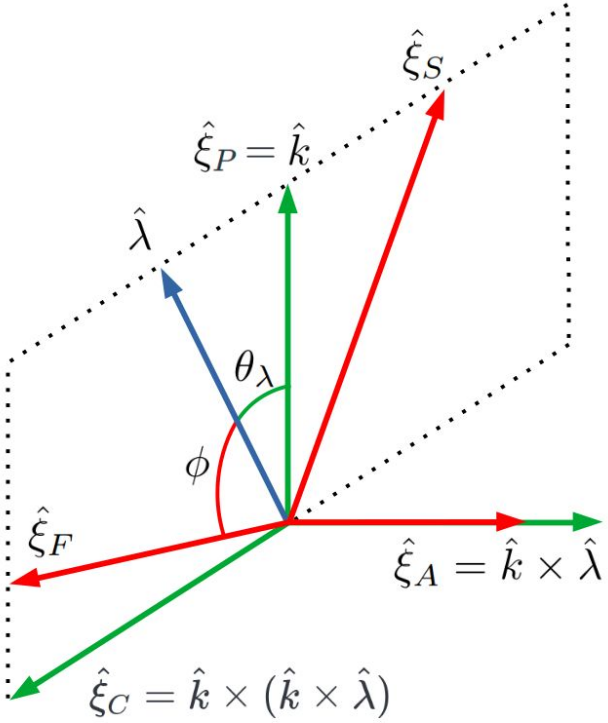
Motivation

Compressibility: a key feature

- Astrophysical plasma is inherently compressible
    - Intermittency & compressions in the solar wind (Hadid+2017, Brodiano+2023, etc.)
    - Density fluctuations in the ISM (Higdon1984, Armstrong+1995, etc.)
- Energy cascades, non-linear interactions, etc. change with compressibility

Motivation

- Kinetic and magnetic fluctuations reside in “wave modes”
  - Transverse (incompressible, “Alfvén” mode)
  - Longitudinal (compressible, “fast” & “slow” modes)



Motivation

Modes couple uniquely to density and magnetic fields

Alfvén mode

- Energy transfer along B-field (Goldreich+1995)
- Particle scattering through gyroresonance (Brandenburg+2013)
- Etc.

Slow mode

- Formation of CNM (Yuen+2001)
- Driving density fluctuation in high-beta (Cho+2002)
- Etc.

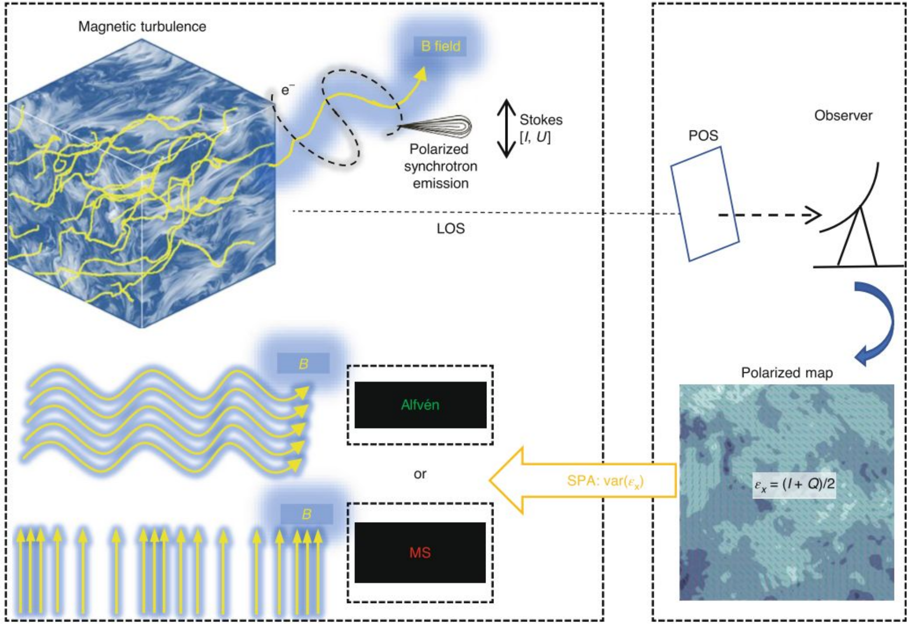
Fast mode

- Isotropic density & pressure fluctuations (Cho+2002)
- Magnetosonic shocks (Elmegreen+2004)
- Particle scattering through TTD (Yan+2002)

The energy fractions of different modes in the ISM have implications for CR transport, star formation, and more,

Method

Can MHD modes be “observed”



Zhang+ (2020)

- No direct observables
- Indirect signatures => Synchrotron Polarization Analysis (SPA) method
- The synchrotron polarization parameter

$$\varepsilon = \frac{I+Q}{2}$$

can provide us information of the dominant mode

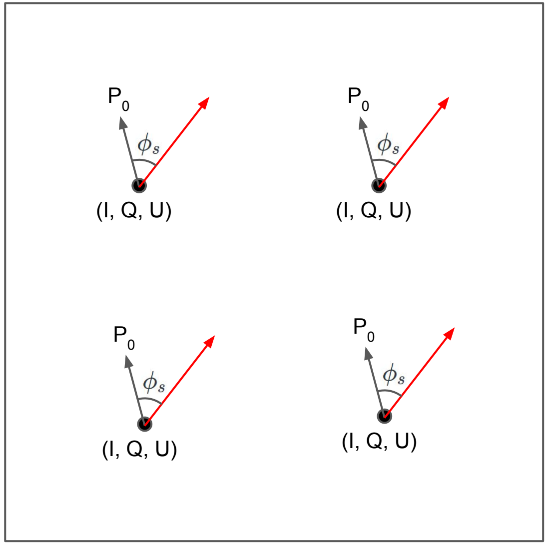
Method

The signature function of SPA

1. Get the mean polarization angle P_0
2. Define an arbitrary vector which makes an angle ϕ_s with P_0
3. Rotate the vector in the range $\phi_s \in (0, \pi)$
4. Compute the "SPA signature" at each ϕ_s

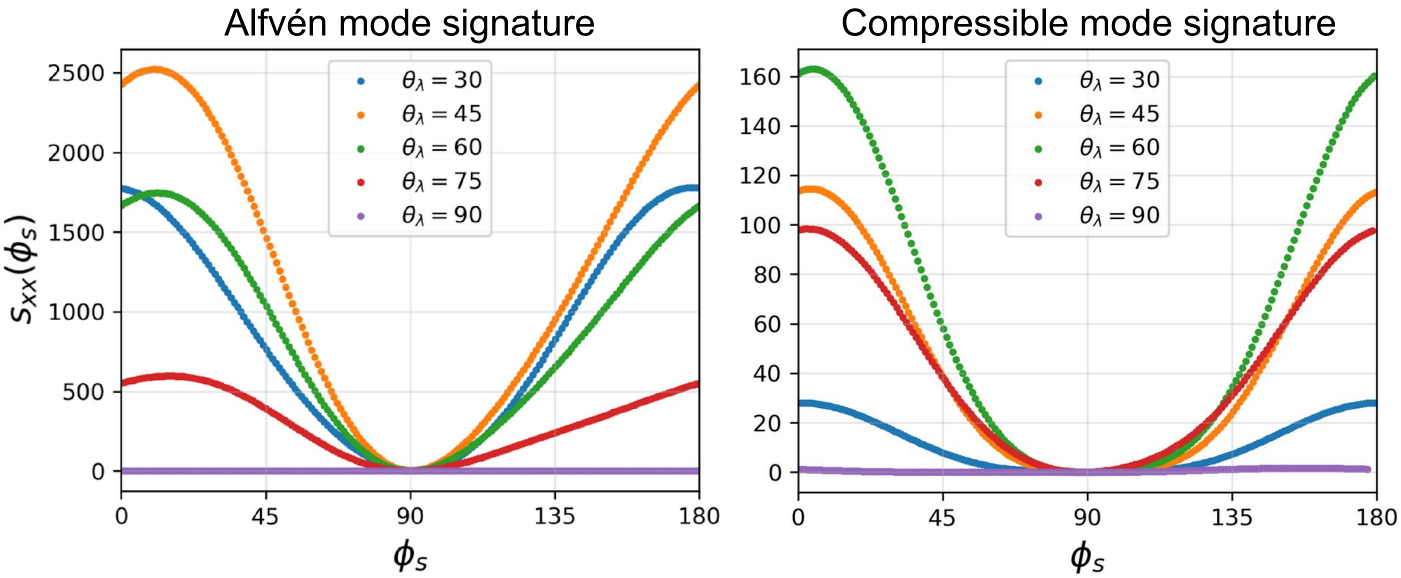
$$s_{xx}(\phi_s) = \mathrm{var}(I + Q_{rot})$$

$$Q_{rot}(\phi_s) = Q\cos\phi_s + U\sin\phi_s$$



Method

The signature function of SPA



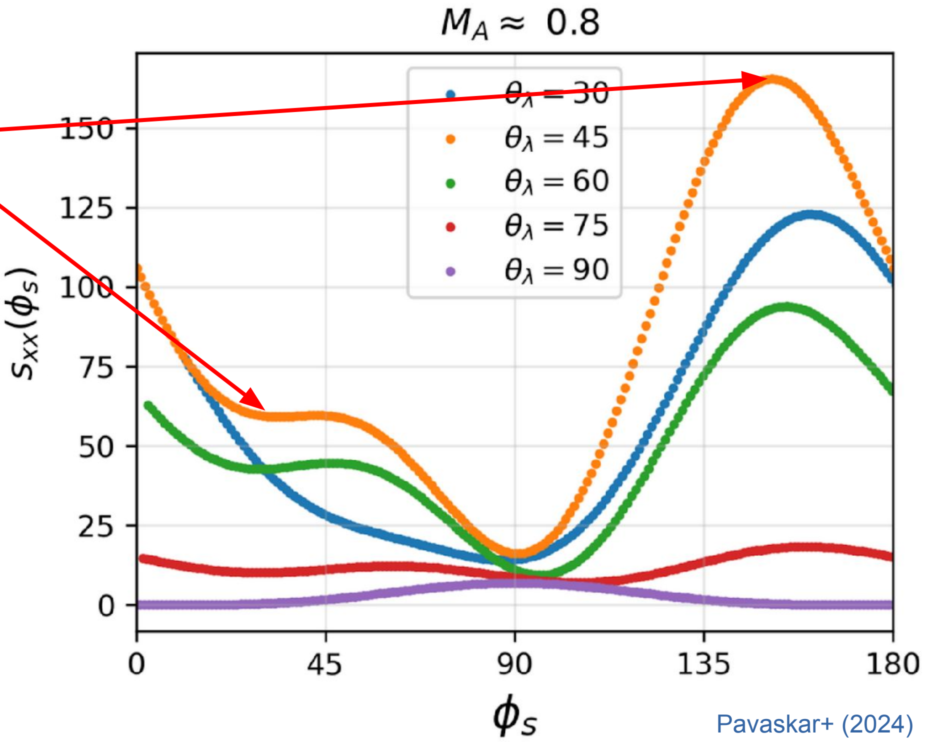
Pavaskar+ (2024)

The concavity of the signature is different for different modes dominating the energy fraction

The signature function of SPA

Asymmetry: Another unique mode feature?

The signatures appear asymmetrical in some cases -> Why?



Pavaskar+ (2024)

Method

The signature function of SPA

We fit the observed signatures to the functional form of s_{xx}

$$s_{xx}(\phi_s) = A_0 + \boxed{A_2}\cos(2\phi_s) + \boxed{A_4}\cos(4\phi_s) + \boxed{B_2}\sin(2\phi_s) + \boxed{B_4}\sin(4\phi_s)$$

Concavity coefficients

Asymmetry coefficients

The ratio $\mathbf{A_4/A_2}$ quantifies the concavity near $\phi_s = \pi/2$

The ratios $\mathbf{B_2/A_2}$ and $\mathbf{B_4/A_2}$ quantify the asymmetry

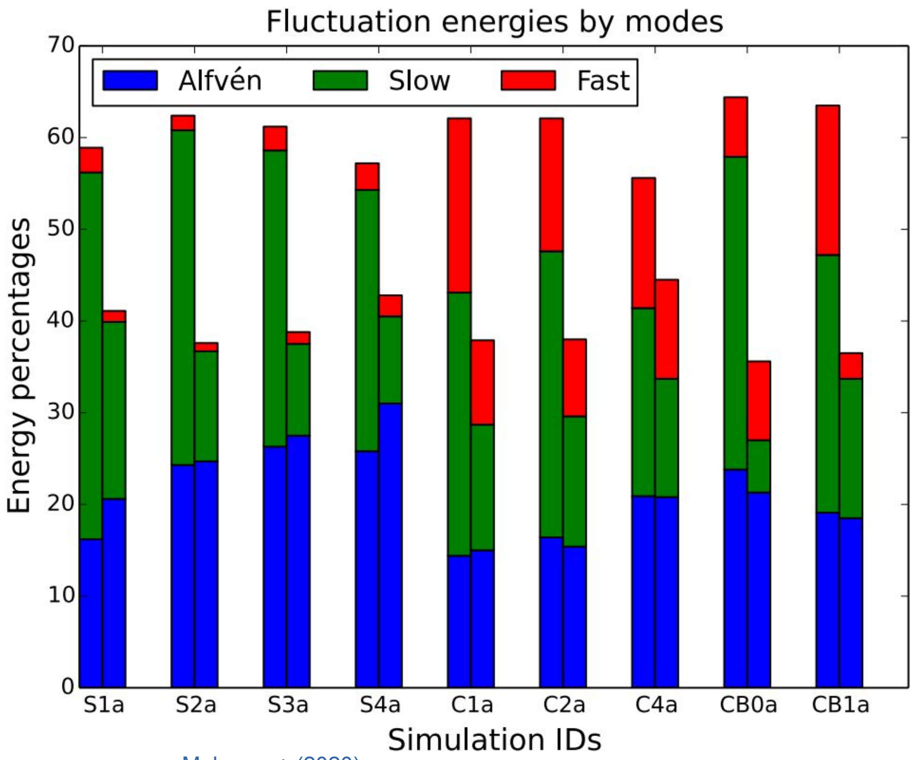
Method

1. Simulate the turbulent ISM with different driving mechanisms

2. Generate synthetic synch. polarization

$$I(\mathbf{R} = (x,y)) = \int dz\left((\mathbf{B}\cdot\hat{x})^2 + (\mathbf{B}\cdot\hat{y})^2\right)$$
$$Q(\mathbf{R} = (x,y)) = \int dz\left((\mathbf{B}\cdot\hat{x})^2 - (\mathbf{B}\cdot\hat{y})^2\right)$$
$$U(\mathbf{R} = (x,y)) = \int dz\left((\mathbf{B}\cdot\hat{x})(\mathbf{B}\cdot\hat{y})\right),$$

3. Test the method and compare with known energy fractions



Makwana+ (2020)

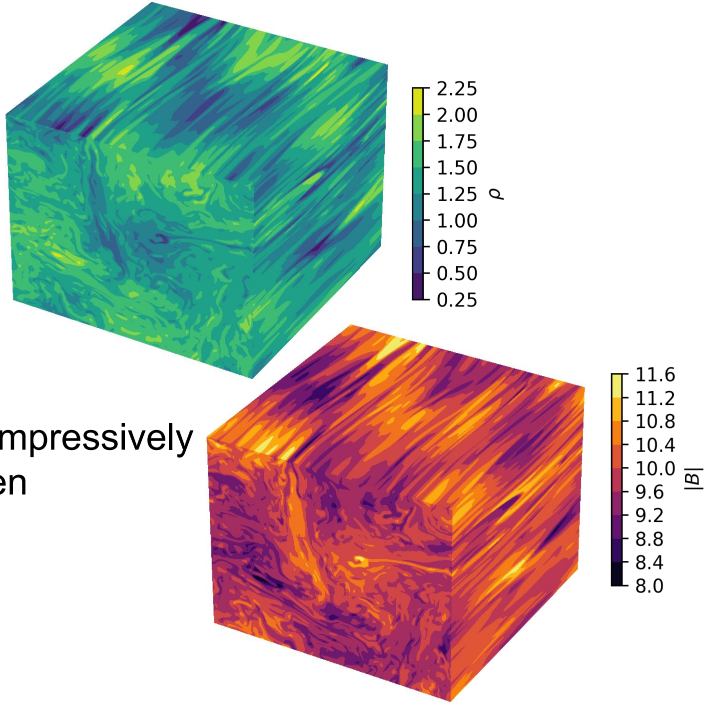
Simulations

Simulating compressible turbulence

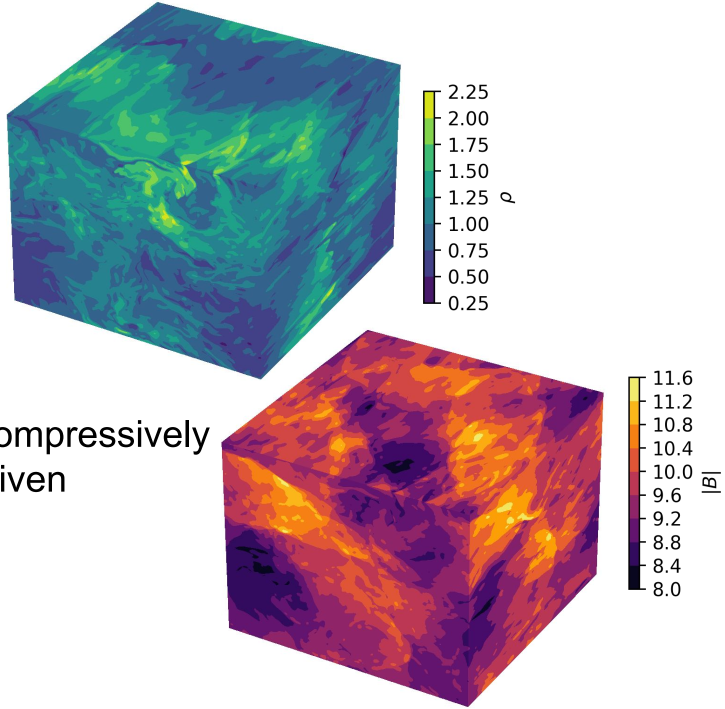
- Generate 3D ideal MHD simulations of plasma turbulence using Athena++
- Isothermal EOS, 3rd order time/spatial integration
- Turbulence evolved for ~5x saturation time
- 16 simulations scanning large parameter range of 2 categories
    a. 8 driven with solenoidal forcing ⇒ containing mostly Alfvén mode
    b. 8 driven with compressive forcing ⇒ containing mostly slow mode with subdominant fast and some Alfvén modes

Simulations

The impact of the driving mechanism



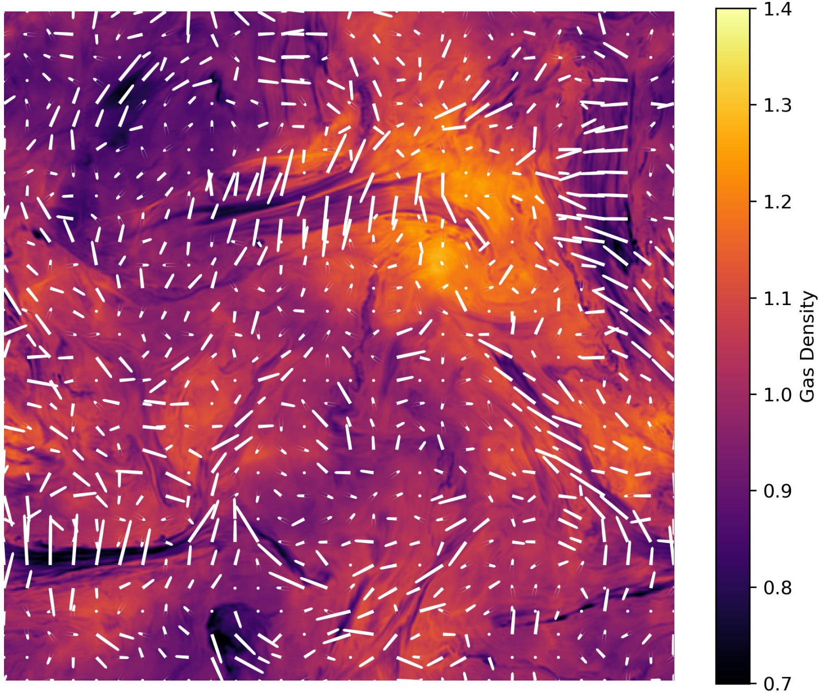
Incompressively driven



Compressively driven

Simulations

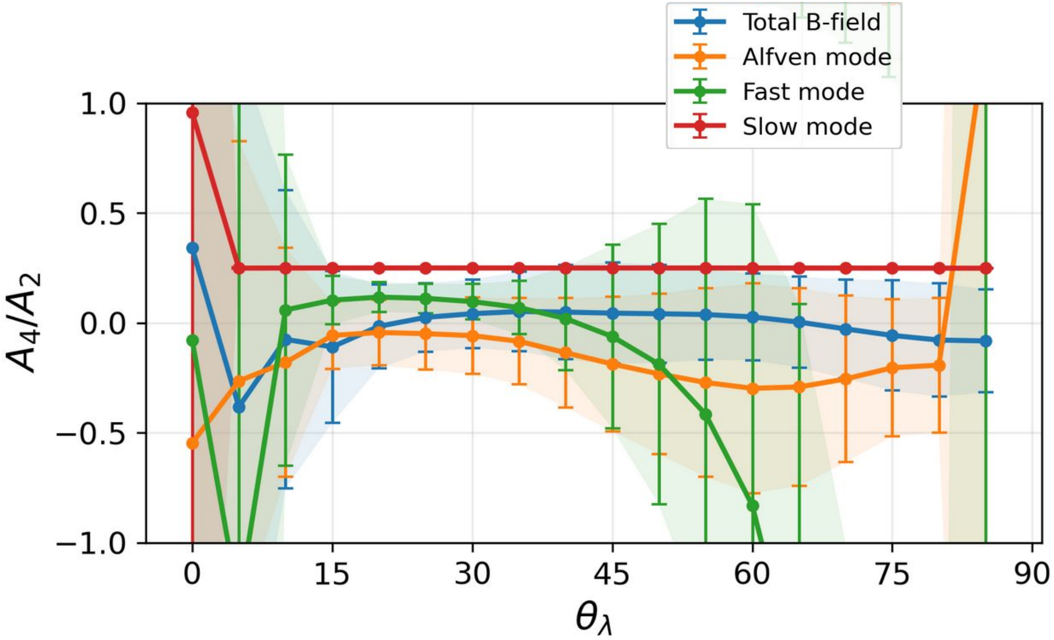
Example of a pol. map from a compressible simulation



Result 1: Identification of A vs C mode dominance

In the <u>incompressible</u> simulations:

- We expect Alfvén turbulence to show a narrow valley in the signature ⇒ $A_4/A_2 < 0$
- The total field follows this behavior within the error ranges

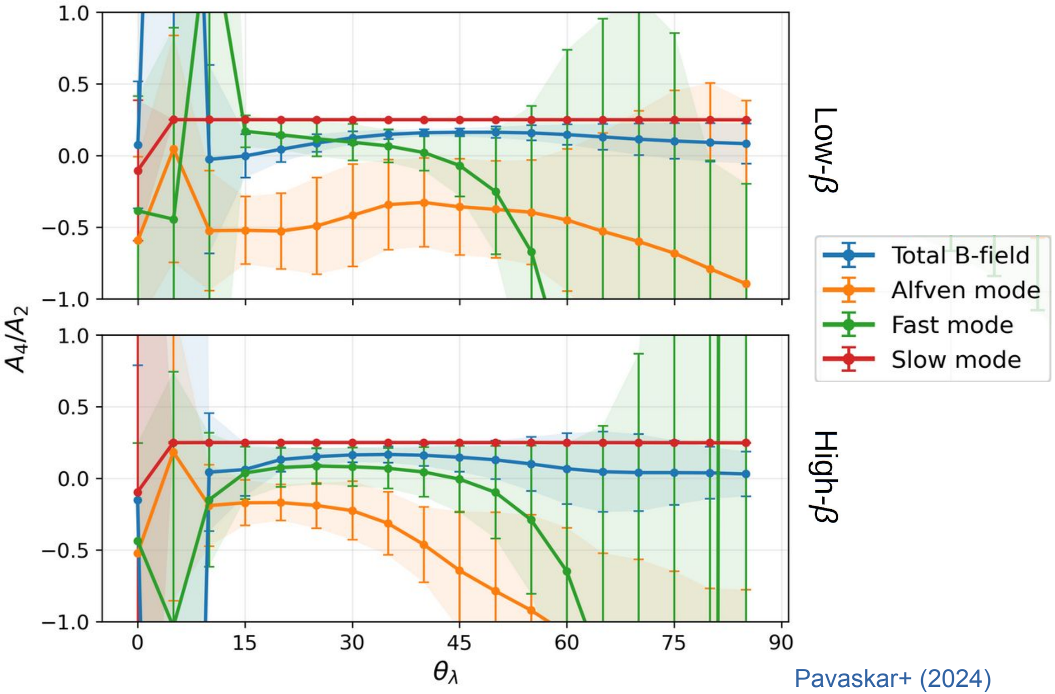


Pavaskar+ (2024)

Result 1: Identification of A vs C mode dominance

In the <u>compressible</u> simulations:

- Flat valley feature expected for compressible/slow mode dominance ⇒ $A_4/A_2 > 0$
- This is seen quite clearly from the total field



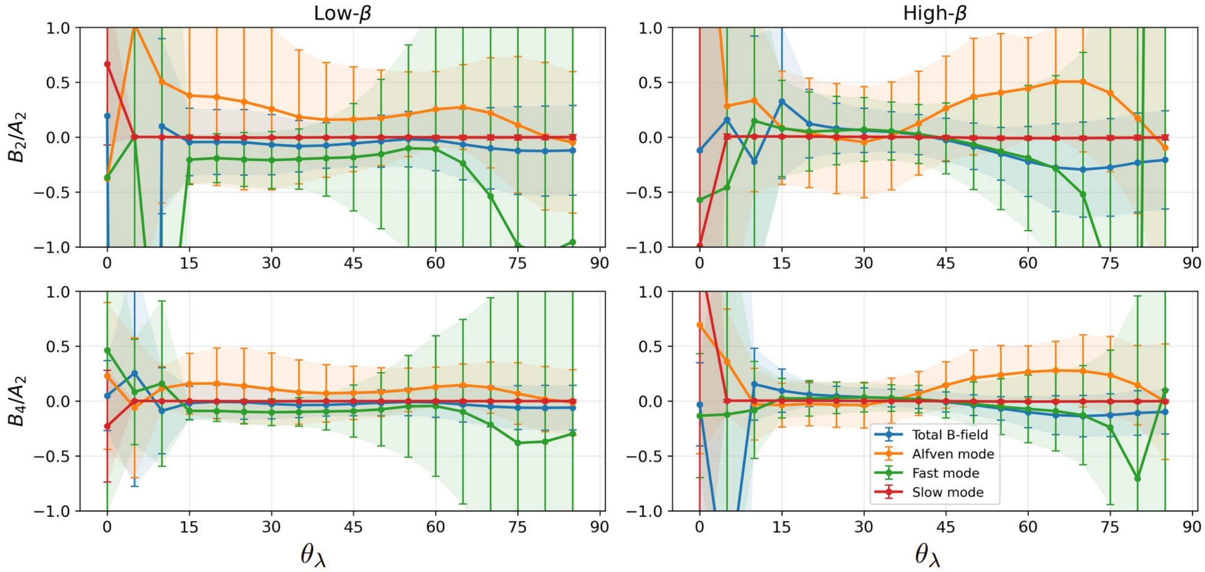
Pavaskar+ (2024)

Result 1: Identification of A vs C mode dominance

This gives us a recipe for distinguishing between Alfvénic and compressible turbulence

1. $A_4/A_2 < 0$ implies Alfvén mode dominates the energy
2. $A_4/A_2 > 0$ implies compressible/slow mode dominates

Result 2: Analysis of signature asymmetry



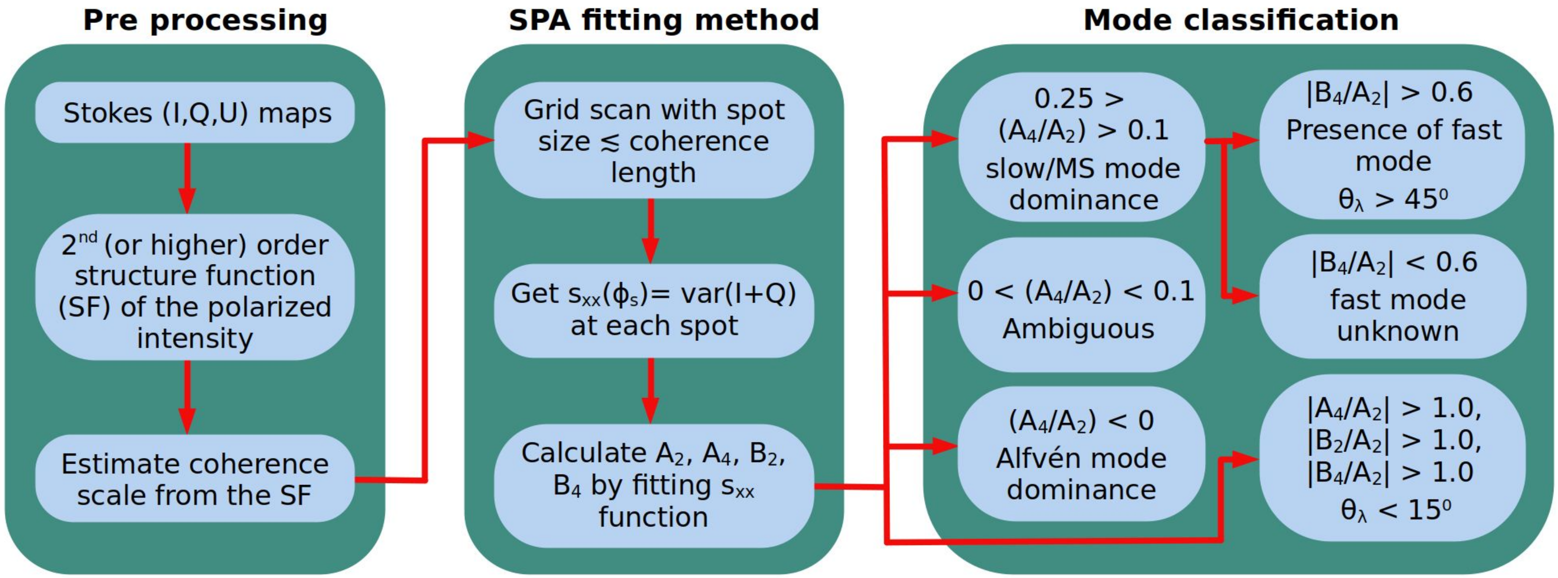
Taking a look at the asymmetry fit parameters

- Slow mode is always symmetric
- At large inclination angles, fast mode shows a large degree of asymmetry

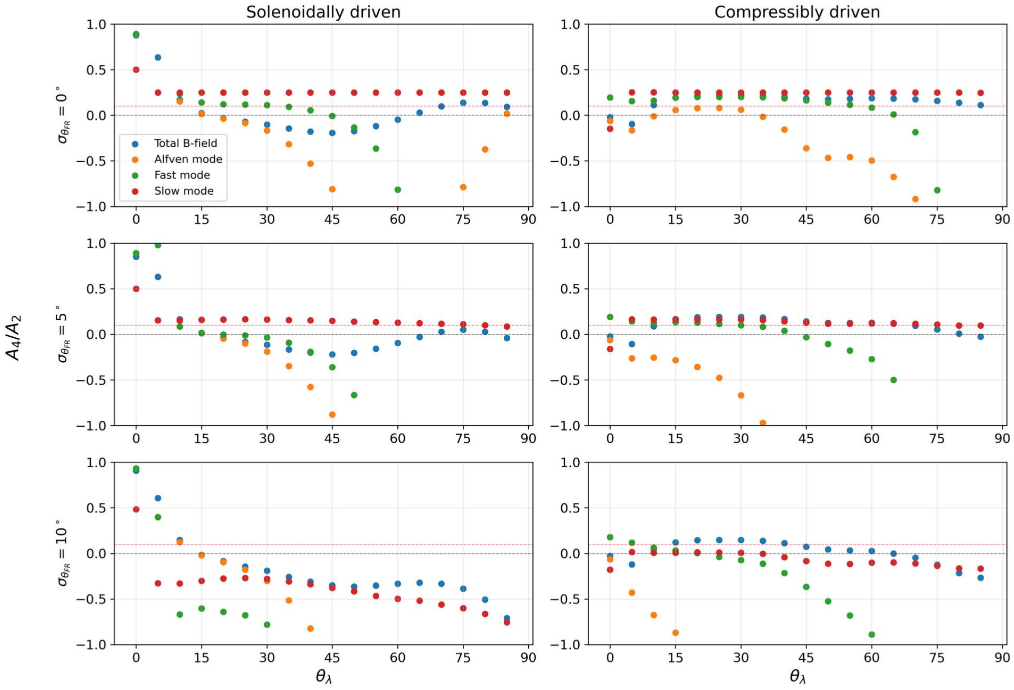
A unique signature of the fast mode predicted for the first time

Pavaskar+ (2024)

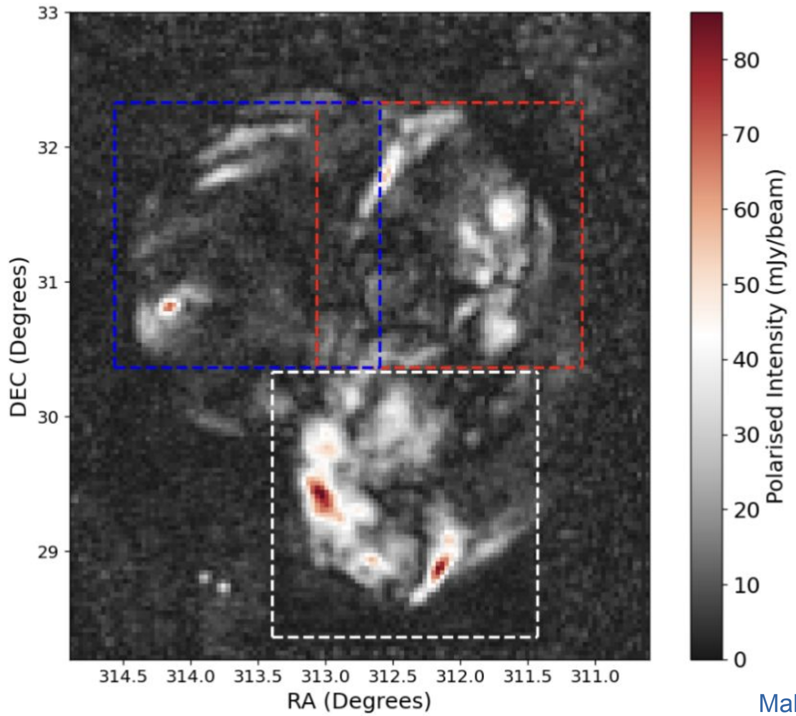
SPA+ in a nutshell

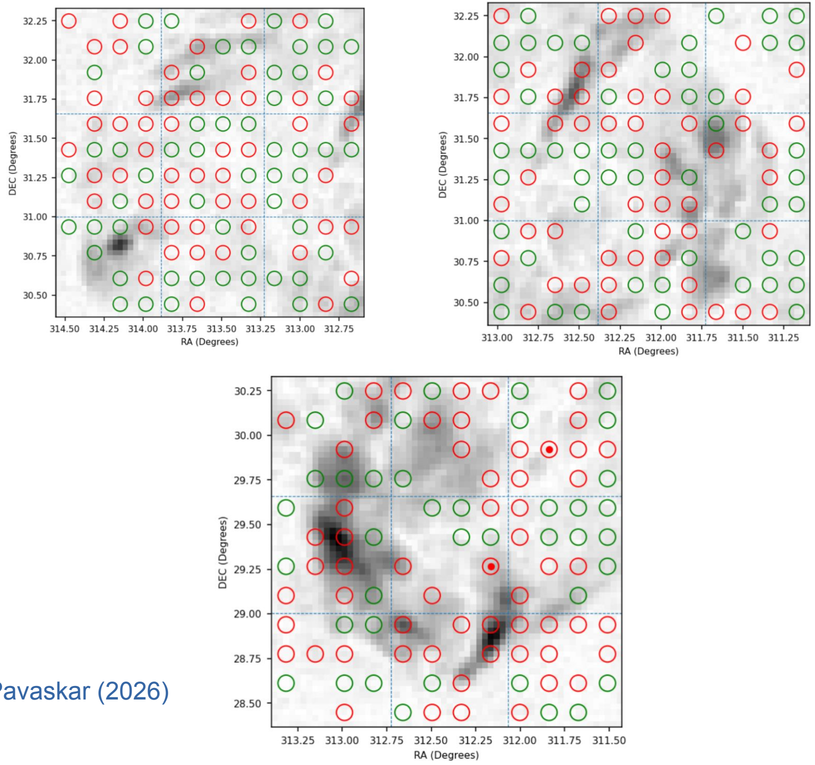


Effect of FR on SPA



Application of SPA





Malik & Pavaskar (2026)

4.3 deg X 4.8 deg Cygnus Loop region

Thank you

Simulating compressible turbulence

- We generate 3D simulations of compressible turbulence using the Athena++ MHD code
- Triply periodic 576^3 simulation box
- Turbulence is driven through an OU process at $k_f < 3$
- Forcing field split into solenoidal (div. free) and compressive (curl free) terms

$\zeta=1$

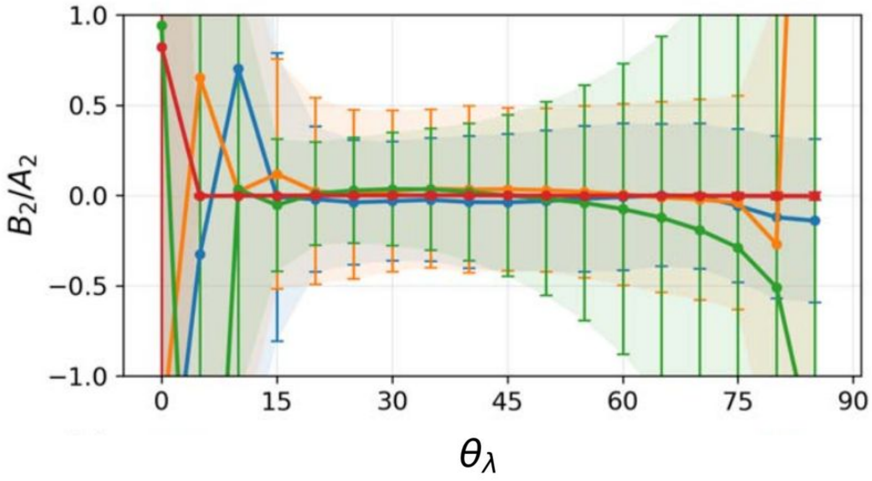
- 8 separate simulations
- Fully incompressible turbulence
- Ranging from
  - sub-Alfvénic to trans-Alfvénic
  - subsonic to supersonic

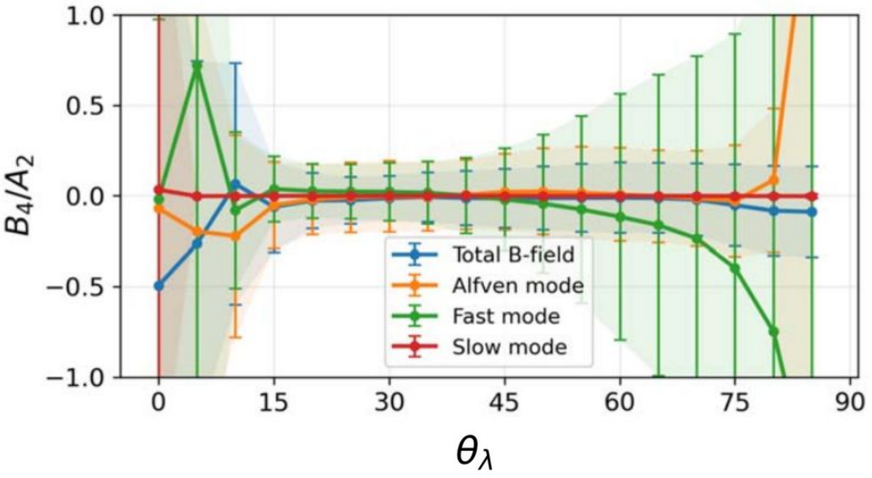
$\zeta=0$

- 8 separate simulations
- Fully compressible turbulence
- Ranging from
  - low plasma-β to high plasma-β
  - Separated into β<2 and β>2

Extra slides

Asymmetry analysis for incomp. simulations





Mode decomposition

Mode basis vectors:

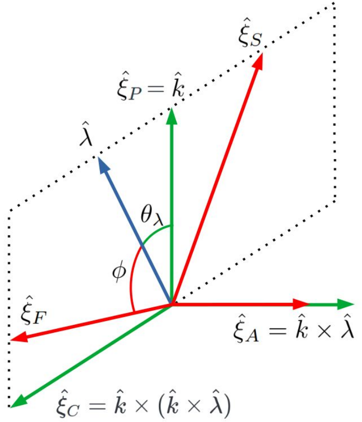
$$\hat{\boldsymbol{\xi}}_A = \hat{\boldsymbol{k}} \times \hat{\boldsymbol{B}}_0 = \hat{\boldsymbol{k}}_\perp \times \hat{\boldsymbol{k}}_\parallel$$

$$\hat{\boldsymbol{\xi}}_s = (-1+\alpha-\sqrt{D})k_\parallel \hat{\boldsymbol{k}}_\parallel + (1+\alpha-\sqrt{D})k_\perp \hat{\boldsymbol{k}}_\perp \qquad \alpha = \frac{\beta}{2}$$

$$\hat{\boldsymbol{\xi}}_f = (-1+\alpha+\sqrt{D})k_\parallel \hat{\boldsymbol{k}}_\parallel + (1+\alpha+\sqrt{D})k_\perp \hat{\boldsymbol{k}}_\perp, \qquad D = (1+\alpha)^2 - 4\alpha\cos^2\theta,$$

Corresponding B field from induction equation

$$\hat{B}(\mathbf{k})_{a,s,f} = \frac{\mathbf{k} \times (\hat{v}(\mathbf{k})_{a,s,f} \times \hat{B}_0)}{\omega_{a,s,f}},$$



Simulating plasma turbulence

Synthetic turbulence

- Variables and fluctuations are constructed statistically
- Assumes spectra, (an)isotropy, (in)homogeneity, etc.
- No dynamical evolution of the plasma

MHD turbulence

- The plasma is described as a fluid, dynamically evolved with a background magnetic field
- Assumes EOS (and non-resistive if ideal)
- Resolves non-linear cascade, anisotropy and shocks
- Does not resolve particle interactions, damping, and other heating-cooling physics

Applicability: Turbulence inertial scale physics

Kinetic turbulence

- Distribution functions of (macro) particles are evolved using Maxwell's equations
- No fluid assumptions
- Resolves ion/electron skin-depth

Applicability: Dissipation scale microphysics / particle interactions